TANNest | Nest report

Generated 2026-11-06 at 18:06 | DRAFT - review before cutting

SAMPLE - supplied BackWall, EndWall and FrontWall DXFs; one copy each. Illustrative prices: $1.00/lb plate, $12.00/ft welding, 25% remnant credit. Default geometry settings and 0.25-inch plate. Not a shop quote.

Estimated total	Stock sheets	Finished seams	Net plate utilization
$6,005.12	8	6 / 32.50 ft	86.7%

Estimate = plate cost - remnant credit + welding. Other fabrication costs are not included.

Verify every sheet in CAM; obtain engineering review of structural or pressure-bearing seams.

Cost and material balance

Item	Quantity	Amount
Plate purchased	5,635.7 lb	$5,635.70
Net parts (including shared-cutline parts)	4,885.4 lb	Included in plate
Usable remnant credit	82.3 lb	-$20.58
Scrap / skeleton	668.0 lb	Included in plate
Welding	32.50 ft	$390.00
ESTIMATED TOTAL		$6,005.12

Stock to purchase

Width x length (in)	Sheets	Plate weight (lb)	Plate cost
72 x 120	2	1,225.2	$1,225.15
72 x 144	6	4,410.5	$4,410.55

Job inputs

Part / source DXF	Qty	Original size (in)	Options
BackWall BackWall.dxf	1	480.500 x 65.000	All features; rotate yes; split yes
EndWall EndWall.dxf	1	116.000 x 66.000	All features; rotate yes; split yes
FrontWall FrontWall.dxf	1	480.500 x 65.000	All features; rotate yes; split yes

Settings and layout comparison

1600 layouts tried in 1.0 seconds. Heuristic search; global optimum not proven.

Setting	Value	Setting	Value
Thickness	0.25 in	Density	0.2836 lb/in^3
Material price	$1.00/lb	Finished weld price	$12.00/ft
Remnant credit	25%	Minimum remnant	12 in
Edge border	0.375 in	Part spacing	1 in
Seam clearance	2 in	Min segment	6 in
Seam rounding	0.0625 in	Avoid notches	True
Arc tolerance	0.005 in	Join tolerance	0.005 in
Search effort	400	Scoring	Cost

Material grade is not stored in the result; thickness and density above are the actual costing inputs.

Alternatives retained by the optimizer

Area is in in squared (drawing units). This is the optimizer's filtered trade-off list, not every attempted layout.

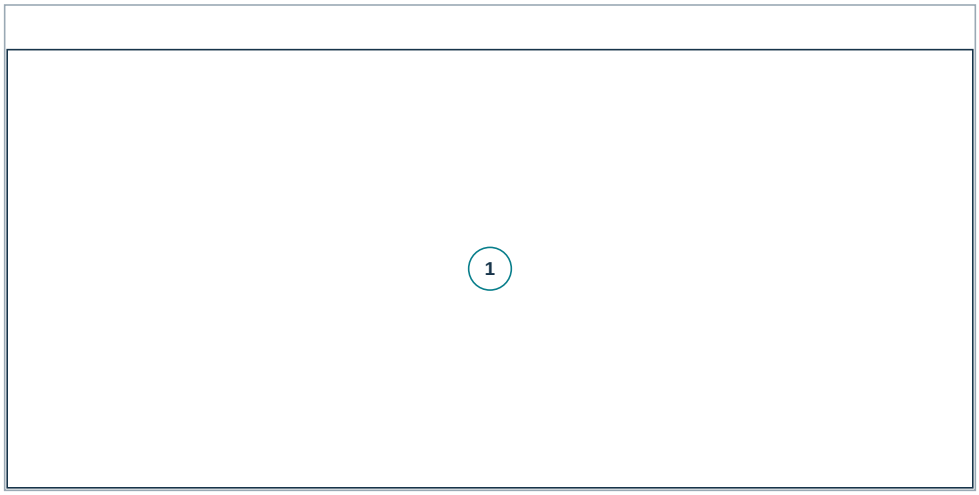
Seams / choice	Sheets	Area (in^2)	Plate	Weld	Credit	Total
6 - chosen	8	79,488	$5,635.70	$390.00	$20.58	$6,005.12
9	8	77,760	$5,513.18	$585.00	$0.00	$6,098.18

Plate-only break-even, 6 to 9 seams: saves 122.5 lb for 16.25 ft more weld (7.54 lb per weld foot; $7.54/ft). This legacy comparison excludes changes in remnant credit; use the totals above for the full comparison.

Sheet 01 | 72 x 144 in

Cut file: Sheet01_72x144_BackWall-S2.dxf

Plate: 735.1 lb | $735.09. Origin: lower left; X along sheet length, Y along width.



Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. BackWall-S2	143.250 x 65.000	0.375 / 0.375	0 deg

Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

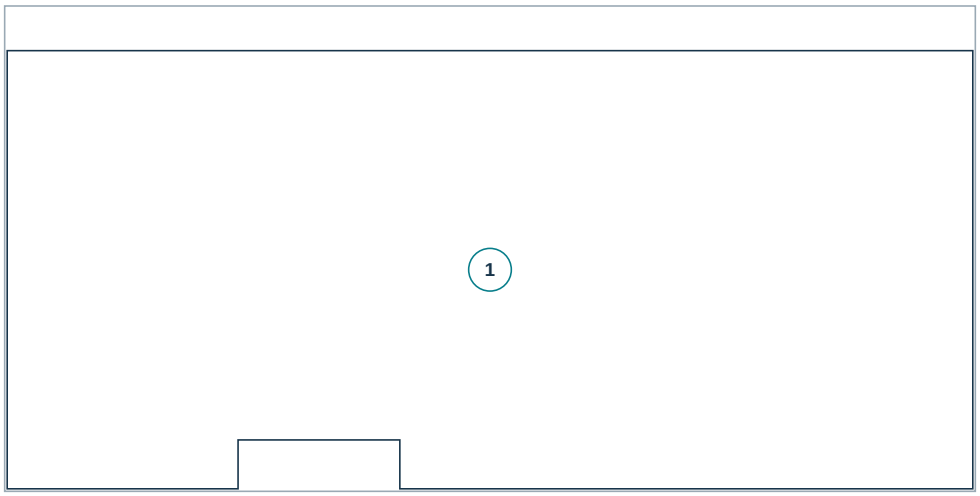
End space past last part: 0.375 in (before spacing allowance; not all end space qualifies for credit).

No strips qualify as usable remnants under the current settings.

Sheet 02 | 72 x 144 in

Cut file: Sheet02_72x144_BackWall-S3.dxf

Plate: 735.1 lb | $735.09. Origin: lower left; X along sheet length, Y along width.



Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. BackWall-S3	143.250 x 65.000	0.375 / 0.375	0 deg

Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

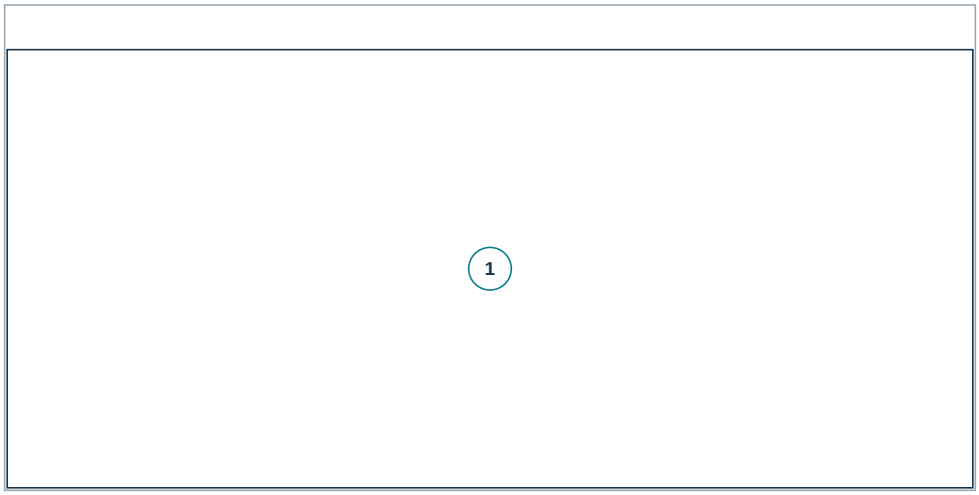
End space past last part: 0.375 in (before spacing allowance; not all end space qualifies for credit).

No strips qualify as usable remnants under the current settings.

Sheet 03 | 72 x 144 in

Cut file: Sheet03_72x144_BackWall-S4.dxf

Plate: 735.1 lb | $735.09. Origin: lower left; X along sheet length, Y along width.



Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. BackWall-S4	143.250 x 65.000	0.375 / 0.375	0 deg

Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

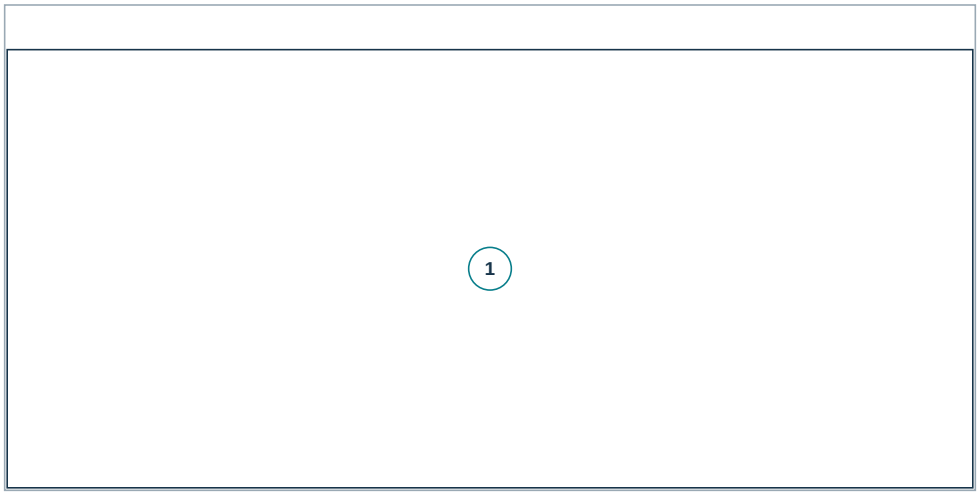
End space past last part: 0.375 in (before spacing allowance; not all end space qualifies for credit).

No strips qualify as usable remnants under the current settings.

Sheet 04 | 72 x 144 in

Cut file: Sheet04_72x144_FrontWall-S1.dxf

Plate: 735.1 lb | $735.09. Origin: lower left; X along sheet length, Y along width.



Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. FrontWall-S1	143.250 x 65.000	0.375 / 0.375	0 deg

Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

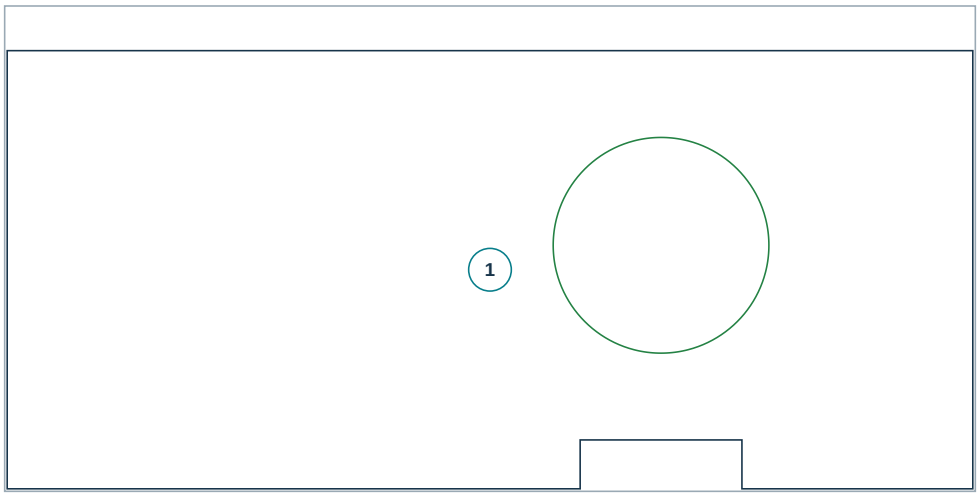
End space past last part: 0.375 in (before spacing allowance; not all end space qualifies for credit).

No strips qualify as usable remnants under the current settings.

Sheet 05 | 72 x 144 in

Cut file: Sheet05_72x144_FrontWall-S2.dxf

Plate: 735.1 lb | $735.09. Origin: lower left; X along sheet length, Y along width.



Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. FrontWall-S2	143.250 x 65.000	0.375 / 0.375	0 deg

Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

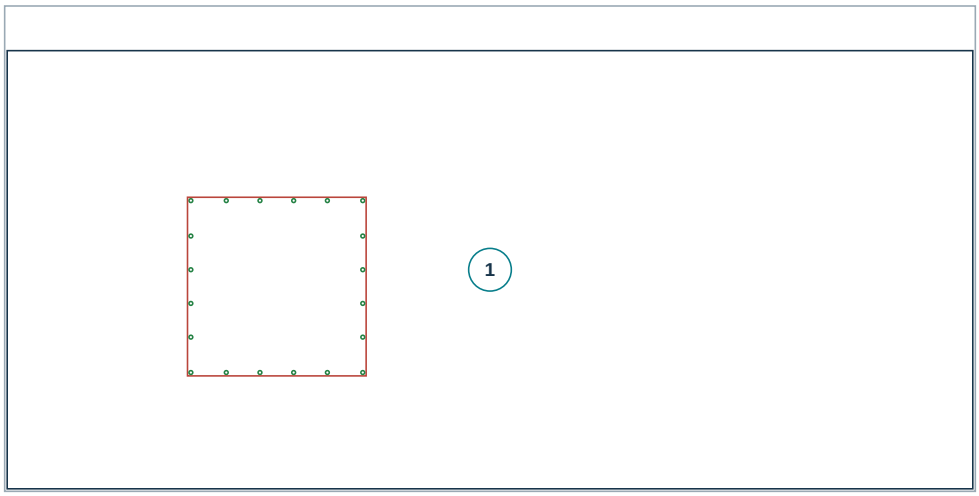
End space past last part: 0.375 in (before spacing allowance; not all end space qualifies for credit).

No strips qualify as usable remnants under the current settings.

Sheet 06 | 72 x 144 in

Cut file: Sheet06_72x144_FrontWall-S3.dxf

Plate: 735.1 lb | $735.09. Origin: lower left; X along sheet length, Y along width.



Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. FrontWall-S3	143.250 x 65.000	0.375 / 0.375	0 deg

Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

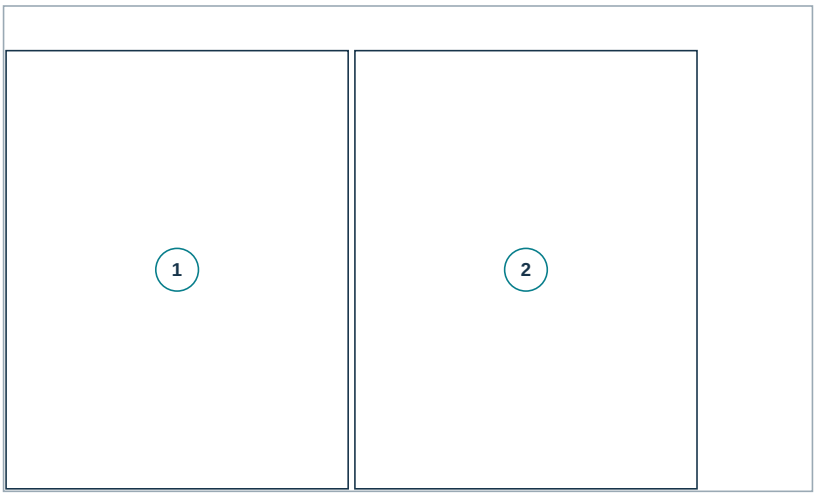
End space past last part: 0.375 in (before spacing allowance; not all end space qualifies for credit).

No strips qualify as usable remnants under the current settings.

Sheet 07 | 72 x 120 in

Cut file: Sheet07_72x120_BackWall-S1+FrontWall-S4.dxf

Plate: 612.6 lb | $612.58. Origin: lower left; X along sheet length, Y along width.



Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. BackWall-S1	50.750 x 65.000	0.375 / 0.375	0 deg
2. FrontWall-S4	50.750 x 65.000	52.125 / 0.375	0 deg

Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

End space past last part: 17.125 in (before spacing allowance; not all end space qualifies for credit).

Usable strip length x width (in)	Weight (lb)	Credit
16.125 x 72.000	82.3	$20.58

Sheet 08 | 72 x 120 in

Cut file: Sheet08_72x120_EndWall.dxf

Plate: 612.6 lb | $612.58. Origin: lower left; X along sheet length, Y along width.

Preview only, not a cutting template. Navy: outside | red: shared cut | green: hole. Numbers match the table.

ID / part	Local size (in)	X / Y (in)	Rotation
1. EndWall	116.000 x 66.000	0.375 / 0.375	0 deg

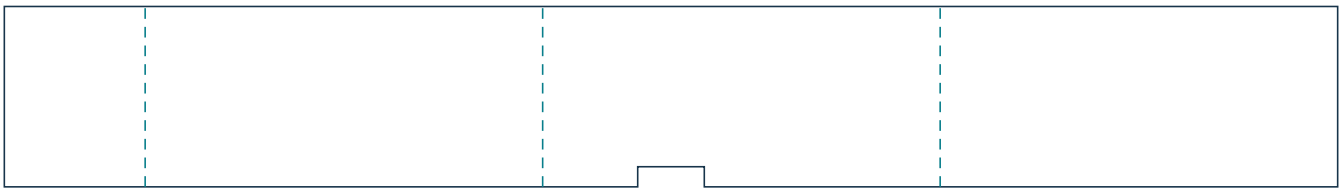
Local size is before sheet rotation. X/Y locates the placed bounding box. Rotation is relative to the normalized part.

End space past last part: 3.625 in (before spacing allowance; not all end space qualifies for credit).

No strips qualify as usable remnants under the current settings.

Wall / part detail | BackWall

Quantity: 1. Coordinates below refer to the original DXF. Normalized long axis: positive original X.



Dashed teal lines show seam positions; preview only. Dimensions and coordinates are in in.

Seam reference file: SeamMap_BackWall.dxf

Seam	Original coordinate	Weld length / copy (in)
1	SEAM X=-189.5000	65.000
2	SEAM X=-46.2500	65.000
3	SEAM X=97.0000	65.000

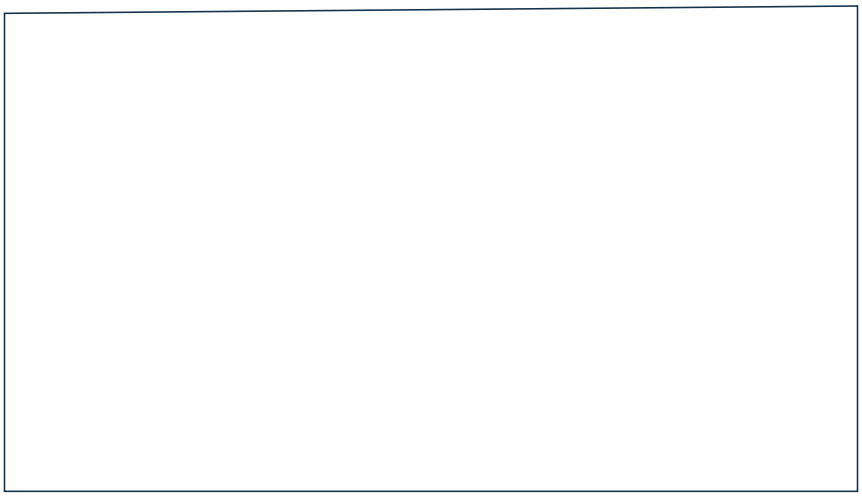
Interval along normalized long axis	Interval length (in)
0.000 to 50.750	50.750
50.750 to 194.000	143.250
194.000 to 337.250	143.250
337.250 to 480.500	143.250

Intervals describe split spans. Irregular outlines can produce multiple pieces within a span; use the actual piece list below.

Actual piece / instance	Sheet	Local piece size (in)
BackWall-S2	01	143.250 x 65.000
BackWall-S3	02	143.250 x 65.000
BackWall-S4	03	143.250 x 65.000
BackWall-S1	07	50.750 x 65.000

Wall / part detail | EndWall

Quantity: 1. Coordinates below refer to the original DXF. Normalized long axis: positive original X.



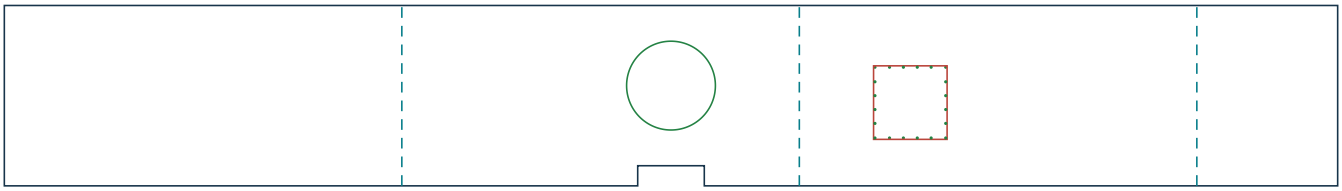
Dashed teal lines show seam positions; preview only. Dimensions and coordinates are in in.

No seams: this part is placed whole.

Actual piece / instance	Sheet	Local piece size (in)
EndWall	08	116.000 x 66.000

Wall / part detail | FrontWall

Quantity: 1. Coordinates below refer to the original DXF. Normalized long axis: positive original X.



Dashed teal lines show seam positions; preview only. Dimensions and coordinates are in in.

Seam reference file: SeamMap_FrontWall.dxf

Seam	Original coordinate	Weld length / copy (in)
1	SEAM X=-97.0000	65.000
2	SEAM X=46.2500	65.000
3	SEAM X=189.5000	65.000

Interval along normalized long axis	Interval length (in)
0.000 to 143.250	143.250
143.250 to 286.500	143.250
286.500 to 429.750	143.250
429.750 to 480.500	50.750

Intervals describe split spans. Irregular outlines can produce multiple pieces within a span; use the actual piece list below.

Actual piece / instance	Sheet	Local piece size (in)
FrontWall-S1	04	143.250 x 65.000
FrontWall-S2	05	143.250 x 65.000
FrontWall-S3	06	143.250 x 65.000
FrontWall-S4	07	50.750 x 65.000